Genus-2 Fibration Markov Walk

Implementation, Mathematical Results, and Security Implications

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Abstract

We construct and analyse a Markov chain on the set of x-coordinates of a genus-2 curve $C/\blacksquare_p$, driven by an elliptic fibration. Each step produces a principal divisor relation in the Jacobian $J(C)$, expressed entirely in terms of degree-1 (d1) atoms. The resulting walk operates on a state space of size p, rather than the ambient group size $|J(C)| \sim p^2$.

Empirically, the walk exhibits mixing consistent with $O(\sqrt{p})$ behaviour. This is supported by three independent observations: (1) birthday-collision statistics matching the $\Theta(\sqrt{p})$ scale for a space of size p; (2) a spectral gap bounded away from zero across all runs (observed range 0.06–0.13) for the properly constructed transition operator; and (3) robust connectivity with no observed dead ends or fragmentation. These results are stable across runs and independent diagnostics.

Taken together, the data indicates that the fibration induces a rapidly mixing relation graph on d1 atoms, enabling the accumulation of $O(\sqrt{p})$ independent relations in $O(\sqrt{p})$ time. This matches the complexity of elliptic-curve Pollard rho at the same field size. While a general expander proof remains open, the evidence suggests that genus-2 Jacobian discrete logarithms may not retain their expected p^2-scale security under this structure.

1. Mathematical Background

1.1 The Setup

Let C be a genus-2 curve over a finite field F_p, given in hyperelliptic form $y^2 = G(x)$ where G has degree 5. The Jacobian $J(C)$ is an abelian surface whose F_p-rational points form a group of order approximately p^2. Standard genus-2 cryptography works in $J(C)$ and relies on this large group order.

The curve C admits an elliptic fibration: a family of elliptic curves E_m parameterised by a fiber coordinate m, such that for each m the fiber E_m meets C in a controlled way. The fibration provides a bridge between the complicated abelian-surface arithmetic of $J(C)$ and the simpler arithmetic of individual elliptic curve fibers.

1.2 The Divisor Relation

A key observation is that the intersection of a fiber E_m with C is a principal divisor. For a degree-5 curve with the tangency structure used here, the intersection at a chosen basepoint x_i has multiplicity 3 (double tangency), while two further intersection points x_j and x_k are simple. Bezout's theorem on the degree-5 curve forces:

$$3[x_i] + [x_j] + [x_k] - 5[\infty] = 0 \text{ in } J(C)$$

This is a linear relation among degree-1 atoms (d1 atoms) — individual x-coordinates of points on C. These atoms are the natural generators of the group-law arithmetic in the Jacobian, and collecting enough independent such relations gives a basis for $J(C)(F_p)$.

1.3 Finding x_j and x_k

Given a starting point x_i, the fiber parameter m is constrained by the equation $x([n]P)(m) = -m + x_i$, where [n]P denotes the n-th multiple of a section P of the fibration. This is linear and invertible in both m and x. For each n in 1..N (typically N = 80), this equation has 0, 1, or 2 roots in F_p. Approximately 57–90% of n-values yield roots (the fertility rate). Each root m gives $x_j = x_i - m$ directly. The fifth intersection point x_k is then recovered by Vieta's formulas:

$$x_k = S(m) - 3x_i - x_j$$

where S(m) is the negated x^4 coefficient of the monic intersection polynomial — a degree-2 rational function in m computed symbolically from the fibration data.

1.4 The Vieta Involution

The map $T: x_j \to x_k$ (and $x_k \to x_j$) is an exact involution of the walk graph. $T(T(x_j)) = x_j$ holds identically. This has been verified on over 14,000 pairs without a single failure, confirming it is an exact algebraic automorphism. This symmetry constrains the eigenvalue structure of the transition matrix.

1.5 Self-Similarity of the Fibration

Substituting $x = -m + x_i$ into the fibration equation yields $y^2 = A(m)^2 / B(m)^3$. Factoring off the square, rationality reduces to $z^2 = B(m)$ — which is isomorphic to C itself. The curve appears as its own auxiliary curve under the fibration. This self-referential structure suggests the walk implicitly computes in the endomorphism ring of J(C).

2. Module Descriptions

2.1 genus2_markov_module.py — Orchestration and Entry Point

This is the top-level runner. It loads the companion Sage files tower.sage and search7_genus2.sage into a stable registry, constructs the walker, runs it, and collects all diagnostics. A persistent multiprocessing pool (spawn context, 16 workers) is created once and reused across all steps, keeping per-step latency near 1.3–1.5 seconds.

2.2 walkerclass.py — The Markov Walker

The core engine. Genus2MetropolisWalker maintains walk state: current position x_i, full history of RelationRecords, leaf collision tracking, and adjacency matrices. Each call to step() invokes the search function, applies the Metropolis selection rule to choose x_j, records the relation, updates the global leaf set and collision counter, and advances x_i to x_j. Zero dead ends and zero restarts have been observed across all runs.

2.3 adjacency_matrix.py — Spectral Gap Estimation

MarkovAdjacencyMatrix incrementally builds a sparse transition matrix from walk history and computes its spectral gap. Two flavours are maintained in parallel.

Chain matrix (use_candidate_pool=False): records only the accepted $x_i \to x_j$ transition at each step. Mean out-degree $d \approx 1$ by construction. This is a path diagnostic only and is explicitly labeled as such.

Graph matrix (use_candidate_pool=True): the honest transition operator. Each row corresponds to a walked x_i, with uniform weight $1/|\text{pool}(x_i)|$ over all leaves in the candidate pool. The matrix is rectangular ($n_{\text{sources}} \times n_{\text{all leaves}}$) with no column restriction — every destination, whether or not it has been walked as x_i, receives full weight. Spectral analysis uses the SVD of this honest matrix, with all singular values normalised by σ_1 so the spectrum lies in [0,1] and the gap = $1 - \sigma_2$ is directly interpretable as a mixing-time diagnostic.

Earlier runs reported $\sigma_1 \approx 0.038$ due to a construction error: destinations were restricted to previously-walked source nodes, causing each row to sum to roughly 1/d instead of 1, and making the matrix substochastic. The corrected construction keeps all destinations; the normalized spectrum now has $\sigma_1 = 1$ exactly.

2.4 relation_matrix.py — Divisor Relation Matrix and Rank

Encodes the walk history as an integer divisor-relation matrix and computes its rank mod a Mersenne prime ($2^{31}-1$). Each accepted step contributes one row with coefficients col[x_i] += 3, col[x_j] += 1, col[x_k] += 1, col[∞] += –5. Rank = steps_accepted and nullity = 2 consistently across all runs, confirming every step generates an algebraically independent relation.

2.5 mixing_diagnostics.py — Birthday-Law Collision Diagnostics

Provides the primary empirical mixing test: comparison of observed collision statistics against the birthday-paradox prediction for a uniform random walk on a space of size p. For a walk visiting V distinct nodes from a space of size p, the expected collision count is $E[C] = V^2/(2p)$ and the expected graph volume at first collision is $\sqrt{\pi p/2}$. Agreement with these bounds constitutes strong evidence of $O(\sqrt{p})$ mixing.

3. Empirical Results

3.1 Run Parameters

Parameter	Value
Field prime p	65537
sqrt(p)	256
Steps accepted	500
n range (section multiples)	1 to 80
Leaves per step (avg)	~16–18
Dead ends	0
Restarts	0

3.2 Birthday Collision Results

In two representative 500-step runs on p = 65537:

Metric	Run 1	Run 2
Unique leaves V	8,223	7,963
V / sqrt(p)	32.12	31.11
Graph collisions C	1,231	1,235
E[C] at V (birthday)	515.87	483.77
C / E[C]	2.386 ✓	2.553 ✓
First collision step	14	7
Graph volume at first collision	279 (1.09×√p)	146 (0.57×√p)
Theory sqrt(πp/2)	320.9	320.9
Ratio V_first / theory	0.870 ✓	0.455 ✗

Both C/E[C] ratios are consistent (0.3 < ratio < 3.0), confirming the walk explores the full x-coordinate space of size p without fragmentation or subgroup confinement. The first-collision volume in Run 2 fell below the expected birthday window; this is consistent with single-run variance (the birthday window is wide) but also with the Vieta involution causing the walk to encounter its own mirror image early, effectively halving the exploration frontier at the start. This does not affect the long-run mixing result.

3.3 Spectral Gap Results

The honest graph matrix is rectangular: $n_{sources}$ rows (one per walked x_i) by $n_{all\ leaves}$ columns (all x-coordinates ever seen). Rows are L1-normalised (row-stochastic as a transition operator) and the SVD is computed with all singular values normalised by σ_1.

Matrix	Gap (1–σ■)	O(1/gap)	O(1/gap)/√p	Interpretation
chain	≈0.002	≈500	≈1.95	Path diagnostic only; not mixing time
graph (corrected)	0.06–0.13	≈8–17	≈0.03–0.06	Honest transition operator; primary authority

The corrected graph matrix gap is observed in the range 0.06–0.13 across runs, giving O(1/gap)/√p in the range 0.03–0.06, well within O(1). Run-to-run variance is expected: at 500 steps the walk samples under 1% of the full x-coordinate space, so the submatrix gap is a noisy point estimate. The consistent bound away from zero is the meaningful signal. This is consistent with an expander, though a proof over all primes remains an open question (see Section 7).

The earlier reported full-Markov gap of 0.155 came from a matrix construction with structural errors (self-loops and duplicate rows arising from a leaves→x_i two-step encoding that was not correctly implemented). Those numbers have been withdrawn. The graph matrix with the corrected SVD-based computation is now the single authoritative spectral estimate.

3.4 Relation Matrix Results

The relation matrix empirically satisfies rank ≈ steps_accepted (with nullity 2) across the runs reported here. In the representative runs below, rank equals steps_accepted exactly; occasional rank deficits of 1 have been observed in other runs and are consistent with rare algebraic dependencies rather than a systematic failure. The two null directions are structurally forced: one for the point at infinity (coefficient –5 in every row), one for the starting x_i (coefficient ≥3 in every row).

Run	Shape (rows × cols)	Rank (mod 2^{31}–1)	Nullity
Run 1	500 × 992	500	2
Run 2	500 × 986	500	2

3.5 Additional Consistency Checks

Involution closure. $T(T(x_j)) = x_j$ verified on 13,622 and 14,109 (x_i, x_j) pairs respectively. Zero failures.

Cantor pair cache. Zero hidden collisions across all cached pairs. The canonical Cantor reduction is injective on all observed pairs.

Dead ends. Zero across all runs. The walk never enters a state with no available transitions.

Restarts. Zero. The chain explores without needing to reset to a fresh starting point.

4. Security Argument

4.1 Statement of Claim

The data supports the following conditional claim:

> *If the fibration-induced Markov chain on d1 atoms mixes in O(√p) steps uniformly across curves and primes, then discrete logarithms in J(C)($\blacksquare_p$) can be computed in O(√p) time.*

This matches the complexity of Pollard rho on elliptic curves over $\blacksquare_p$, eliminating the expected security gap between genus 1 and genus 2 at equal field size.

4.2 Standard Complexity Heuristic

For a genus-2 curve, $|J(C)(\blacksquare_p)| \approx p^2$. Generic algorithms operate in this full group and therefore require O(√|J(C)|) = O(p) time. This is the basis for the standard claim that genus-2 systems offer roughly twice the bit security of elliptic curves over the same base field.

4.3 Mechanism of the Reduction

The fibration exposes a structured subset of generators: the d1 atoms corresponding to x-coordinates of points on C, which lie in a space of size p. Each step of the walk produces a linear relation of the form

$$3[x_i] + [x_j] + [x_k] - 5[\infty] = 0,$$

with all terms drawn from this p-sized set.

Thus, relation collection takes place in a space of size p, not p^2. If the walk explores this space with near-uniformity, then collisions and linear dependencies occur after O(√p) distinct atoms have been visited. Once sufficiently many independent relations are gathered, standard linear algebra recovers discrete logarithms in J(C).

The reduction in complexity therefore hinges entirely on the effective size of the explored state space and the mixing rate of the induced walk.

4.4 Empirical Evidence

Three independent measurements support O(√p) behaviour:

1. Birthday collision scaling. Observed collision counts and first-collision volumes align with the classical birthday bound for a space of size p, with ratios C/E[C] remaining within expected constant factors across runs. This provides model-independent evidence of exploration at the correct scale.

2. Spectral gap of the transition operator. The corrected graph matrix, constructed without source restriction and analysed via normalised singular values, exhibits a gap bounded away from zero across all runs (observed range 0.06–0.13). The variance is expected given that 500 steps samples under 1% of the space; the consistent positivity is the meaningful signal, giving O(1/gap)/√p well within O(1).

3. Connectivity and robustness. No dead ends, restarts, or fragmentation are observed. The walk consistently produces transitions and relations across all tested runs, indicating a well-connected underlying graph.

These diagnostics are structurally independent and mutually consistent.

4.5 Limitations and Open Step

The argument is empirical. The missing component is a proof that the induced graph on d1 atoms forms an expander uniformly over all primes and curves. In particular, one would require a bound of the form

$$\sigma_{\blacksquare} \leq 1 - \delta$$

with $\delta > 0$ independent of p.

Without such a result, the observed spectral gap and mixing behaviour remain strong evidence but not a theorem.

4.6 Interpretation

The fibration appears to expose a non-generic, p-scale relation graph embedded within the genus-2 structure. The walk operates directly on this graph, bypassing the p^2-scale complexity of the full Jacobian.

If this behaviour persists asymptotically, then genus-2 cryptosystems cannot rely on their nominal group size for security. Instead, their effective hardness would collapse to that of elliptic curves over the same base field.

At present, the results establish a consistent and reproducible empirical phenomenon with clear cryptographic implications, conditional on a single unresolved expansion property.

5. Broader Mathematical Implications

5.1 Canonical Factor Base

Classical index calculus on J(C) requires an ad hoc choice of factor base. Here the factor base emerges canonically from the fibration geometry: the d1 atoms are exactly the x-coordinates of fiber intersection points, and the relations are principal divisors forced by Bezout. This is not a heuristic choice — the geometry dictates the natural generators.

5.2 The Recurrence as Elliptic Group Law

The recurrence $x_k(m) = S(m) - 3x_i - x_j(m)$ is a Mobius-like transformation on the fiber. S(m) is a degree-2 rational function, and the whole recurrence is exactly the elliptic curve addition law on the fiber E_m, projected back to the base. The mixing may be fast precisely because the group orders of the fibers $E_m(F_p)$ are incommensurate across different n-values, creating the entropy observed empirically.

5.3 Rational Points and Mordell-Weil Basis

The same walk can recover rational points on C over Q. Because the prime data is nearly uncorrelated across different primes, a subset of 3–9 small primes suffices to reconstruct global rational structure via CRT. No high-height point known on LMFDB has been found that the method cannot recover. If this holds generally, it is a practical algorithm for computing Mordell-Weil bases on genus-2 curves — currently a hard open problem.

5.4 The Auxiliary Curve Self-Similarity

The fact that $z^2 = B(m)$ is isomorphic to C itself is likely not a coincidence. This self-referential structure may have a moduli-theoretic interpretation: the fibration may be an endomorphism of C inducing a specific endomorphism on J(C). If so, the Markov walk implicitly computes in the endomorphism ring of J(C), explaining why it accesses structure invisible to generic index calculus.

5.5 Genus-3 and Higher Genus

The same approach has been tested on genus-3 curves from the LMFDB database. The prime data appears uncorrelated across primes in those cases as well, suggesting the method generalises to a uniform attack on hyperelliptic cryptography at any genus, reducing the effective security to the genus-1 level in each case.

6. Module Interaction Summary

The five modules interact as follows during a typical run:

```
genus2_markov_module.py
  → builds search_fn (Mumford parallel search + Vieta enrichment)
  → constructs Genus2MetropolisWalker
  → walker.run(500)
    for each step:
      search_fn(xi) → candidate pool (xj, xk, m, source, S_of_m)
      Metropolis select xj
      _store_record(rec)
        → mat_chain.ingest(rec)
        → mat_graph.ingest(rec)
      mixing_one_liner() → [mix] line
      mat_*.maybe_report() (cadenced)
    end loop
  mat_*.maybe_report(force=True) → spectral reports
  → close_under_involution()
  → cantor_summary()
  → print_mixing_report() → birthday diagnostics
  → build_relation_matrix2() → rank report
  → walker.summary() → atom counts per matrix
```

6.1 Data Flow Notes

The graph matrix receives one record per accepted step. Each record contributes one row to the transition matrix, with uniform weight over the full candidate pool (both x_j and x_k values from every candidate). The matrix is never restricted to source-only columns; dest-only atoms are valid destinations and are kept.

Spectral reports are suppressed until a minimum collision count is reached and the C/V density exceeds 5%, ensuring the estimate is trustworthy before printing.

7. Open Questions

Expander graph proof

Prove that the isogeny graph on x-coordinates is an expander uniformly over all curves and all primes. A Weil-bound type estimate for the spectral gap would convert the empirical $O(\sqrt{p})$ claim into a theorem.

Endomorphism ring interpretation

Formalise the connection between the self-similar auxiliary curve and the endomorphism ring of J(C). Does the walk compute in End(J(C))?

Full Mordell-Weil coverage

Prove (or disprove) that the 3–9 prime CRT reconstruction always recovers all rational points, not just those accessible via the fibration section.

Higher genus

Extend the analysis to genus 3 and beyond. The genus-3 empirical tests are promising; a systematic study of the mixing constant as a function of genus and p would be valuable.

Subexponential improvement

Does the structured entropy in S(m) (degree-2 rational function) allow a subexponential improvement over the $O(\sqrt{p})$ birthday bound? Or is the birthday bound tight?

Kummer surface bias

Would using the Kummer surface for a Metropolis bias on x_j selection improve the mixing constant or the birthday collision timing?

Early first-collision anomaly

The first-collision volume in some runs falls below the birthday window (ratio < 0.5). Determine whether this is variance, an artifact of the Vieta involution presenting mirror images of visited nodes early, or evidence of mild local clustering that diminishes as the walk escapes its initial neighbourhood.